



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



FULL STACK DEVELOPMENT

AN INTERNSHIP REPORT

Submitted by

K KEERTHI – 20221ISE0046

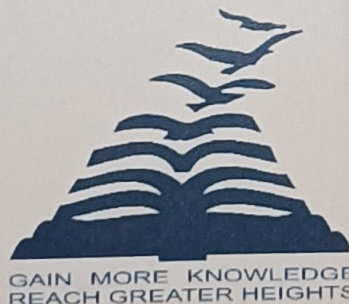
Under the guidance of,

Mr. Sreehari T M

BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND ENGINEERING



PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026



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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report “**FULL STACK DEVELOPMENT**” is a bonafide work of **K KEERTHI (202211SE0046)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in **INFORMATION SCIENCE AND ENGINEERING** during 2025-2026.

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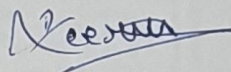
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DECLARATION

I am a student of final year B.Tech in INFORMATION SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named **K KEERTHI**, hereby declare that the internship work titled “**Full Stack Development**” has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in INFORMATION SCIENCE ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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DATE: 28/04/26

INTERNSHIP COMPLETION CERTIFICATE



रेल पहिया कारखाना RAIL WHEEL FACTORY

भारत सरकार / Govt. of India
रेल मंत्रालय / Ministry of Railways

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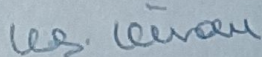
DATE: 17.04.2026

CERTIFICATE

This is to certify that Ms. K KEERTHI, (2022HSE0046), B.Tech (Information Science & Engineering) Student of Presidency University, has underwent Internship on "Full Stack Web Development and Real-Time Multilingual Speech to Text Translation System" at Rail Wheel Factory, Yelahanka, Bangalore for a period of 03 months i.e., 05th January 2026 to 15th April 2026.

The student has shown keen interest in learning during the period she was with us for the Internship. We found her Enthusiastic, Diligent, Hard Working and Creative.

We wish a bright future and success in all her endeavors.


(K.S.KIRAN)

SENIOR PERSONNEL OFFICER-I

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Mr. Sreehari T M, Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, for his moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Gopal Krishna Shyam, Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

I express my cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS, **Dr. Shakkeera L**, Associate Dean, Presidency School of Computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

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I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

K KEERTHI

ABSTRACT

This internship was carried out in the domain of Full Stack Development at Rail Wheel Factory (RWF), Bengaluru. The primary objective of the internship was to design and develop a web-based manufacturing management system to facilitate efficient report generation, improve data accessibility, and optimize workflow performance in an industrial environment. The work involved developing APIs, integrating databases, and building responsive frontend interfaces for real-time report visualization.

During the internship, SQL Server was used for database operations, Fast API was used for backend development, and ReactJS was used for frontend implementation. The backend APIs were implemented using asynchronous programming (async/await), enabling non-blocking execution and efficient handling of multiple requests. This significantly improved system performance compared to traditional synchronous approaches.

The system also included processing of PDF reports provided by the organization, which were converted into HTML format to enable structured data extraction and seamless integration into the web application. The frontend dynamically rendered processed report data using asynchronous API calls, ensuring real-time updates without requiring page reload.

In addition to the core internship work, a mini-project titled “Full Stack Development and Real-Time Multilingual Speech-to-Text Translation System” was developed. This system captures live audio input through a microphone, performs speech recognition to convert Telugu speech into text using advanced Automatic Speech Recognition (ASR) models, and further translates it into English using multilingual translation models. The translated output is displayed on the frontend, demonstrating a real-time multilingual communication pipeline.

Overall, the internship resulted in the successful development of a complete full-stack reporting system along with an AI-based speech translation application, enhancing both technical expertise and practical understanding of modern web and intelligent software systems.

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ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
ASR	Automatic Speech Recognition
SQL	Structured Query Language
UI	User Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
HTTP	Hypertext Transfer Protocol
CRUD	Create, Read, Update, Delete
RWF	Rail Wheel Factory
HTML	Hypertext Markup Language
PDF	Portable Document Format
JS	JavaScript
TS	TypeScript
DBMS	Database Management System
VS Code	Visual Studio Code
URL	Uniform Resource Locator
AI	Artificial Intelligence
NLP	Natural Language Processing
NLLB	No Language Left Behind
PCM	Pulse Code Modulation

Chapter 1

INTRODUCTION

1.1 Background

Rail Wheel Factory (RWF), Bengaluru, is one of the major production units of Indian Railways, responsible for manufacturing wheels, axles, and wheelsets used in railway transportation systems across India. Due to the large volume of manufacturing data generated daily, efficient software systems are required for accurate reporting, record management, and workflow optimization.

In modern industrial environments, full-stack web development plays a crucial role in digitizing manual processes and improving system efficiency. Web-based applications enable centralized data storage, real-time report generation, and seamless access to information across departments.

This internship was carried out in the domain of Full Stack Development at Rail Wheel Factory, where the primary focus was on designing and developing manufacturing management reporting systems using SQL Server, Fast API for backend REST APIs, and ReactJS for frontend implementation.

In addition to the core internship work, a mini-project on real-time multilingual speech-to-text translation was also developed. This project supports multiple Indian languages including Telugu, Kannada, Hindi, and Tamil. It demonstrates the integration of speech recognition and machine translation technologies to enable real-time conversion of speech into English text, thereby enhancing multilingual communication capabilities.

1.2 Internship Statistics

The internship was undertaken in the 2026 academic year as a part of the CSE7302 Internship course requirement at Presidency University.

Internship Details

Name of the organization: Rail Wheel Factory (RWF).

Location: Yelahanka, Bengaluru, Karnataka.

Presidency School of Computer Science and Engineering, Presidency University.

Domain: Full Stack Development (Internship)

Duration: Semester Internship (5th Jan – 15th April)

Technologies Used: SQL Server, Fast API, ReactJS, JavaScript, TypeScript, Postman, GitHub.

Mini Project Domain: AI-based Multilingual Speech Recognition and Translation.

1.3 Existing System

In the pre-modernization period, most manufacturing reports within the industrial setting were done manually or with semi-digital obsolete systems. There were several limitations of these systems: Delay in generation of reports, Repetitive manual work, Inability to have real-time data, Disagreement in management, Increased possibility of human errors, Reduced scalability in case of upgrades. The conventional PDF reports were dynamic and hard to interact with dynamically in web applications.

1.4 Proposed Approach

The proposed system is aimed at developing a fully web-based reporting system of Rail Wheel Factory.

The approach includes

1. Writing Fast API-based backend REST APIs.
2. Linking APIs and SQL Server database.
3. Turning synchronous APIs to asynchronous APIs to achieve high performance.
4. PDF report processing by converting to HTML.
5. Parsing HTML files to get structured data.
6. Displaying data dynamically in ReactJS frontend
7. Adopting real-time frontend rendering with asynchronous fetches.

Additionally, the mini-project follows this approach:

- Capturing real-time audio input using a microphone
- Performing multilingual speech recognition (Telugu, Kannada, Hindi, Tamil) using ASR models
- Translating recognized text into English using multilingual translation models
- Displaying translated output dynamically on the frontend

1.5 Objectives of Internship

The main objectives of the internship were:

- To gain insight into industrial software systems in the real world.
- To create full-stack web applications in manufacturing sphere.
- To develop REST interfaces to generate reports.
- To make use of asynchronous back-end processing.
- To dynamically integrate frontend with backend.
- To enhance the reporting performance at RWF.
- To develop a real-time multilingual speech-to-text translation system

1.6 SDGs Addressed

This internship will help in the following Sustainable Development Goals (SDGs):

SDG 8 - Decent Work and Economic Growth. Enhances industrial efficiency via programs automation.

Goal 9: Industry, Innovation, and Infrastructure. Encourages technological advancement in the production processes.

SDG 12 - Responsible Consumption and Production. Allows effective electronic workflows that decrease paper work.

1.7 Introduction to Internship Report.

The present report provides the full internship experience in Rail Wheel Factory.

Chapter 1 provides the background and objectives.

Chapter 2 will review related technologies and literature.

Chapter 3 describes the stages of methodology and implementation.

Chapter 4 is about project planning and project management.

Chapter 5 introduces system analysis and design.

Chapter 6 describes the tools, hardware, and software used.

Chapter 7 Assesses findings and testing results.

Chapter 8 deals with the ethical and sustainability.

Chapter 9 is the final chapter of the internship findings.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

Modern industrial software systems rely heavily on full-stack web technologies, efficient backend architectures, and intelligent data processing techniques. In recent years, industries have transitioned from manual and legacy systems to web-based applications that support real-time data accessibility, scalability, and automation.

This chapter reviews the key technologies and concepts related to full-stack development, REST API architecture, asynchronous processing, frontend frameworks, PDF data processing, and AI-based multilingual speech translation systems. These technologies form the foundation of the work carried out during the internship.

2.2 Full Stack Web Development in Industrial Systems

Full stack development refers to the integration of frontend, backend, and database technologies to build complete web applications. In industrial environments, such systems are used to manage production data, employee records, and operational workflows.

Studies indicate that full-stack systems improve efficiency by enabling centralized data management and reducing manual intervention. Technologies such as SQL databases, backend frameworks like FastAPI, and modern frontend libraries such as ReactJS allow industries to build scalable, maintainable, and high-performance applications.

2.3 REST API Architecture

Representational State Transfer (REST) is a popular architectural language to design web services. REST APIs allow an interaction between front and back-end systems based on the HTTP methodology (GET, POST, PUT and DELETE).

RESTful APIs have several benefits

- Frontend / backend separation of concerns.
- Scalability and flexibility
- Simple incorporation with various customers.

- Normalized communication in the form of JSON.

REST APIs to process the manufacturing report data were written in Flask in this internship.

2.4 Asynchronous API Processing

Conventional synchronous APIs are designed to handle only a single request at a time and this may cause delays and wastage of resources. On the other hand, asynchronous APIs can be executed non-blockingly, thus enabling multiple requests to be processed concurrently.

Studies have shown that asynchronous processing enhances the performance of the system particularly application processes that involve a large amount of data or real time data. in this project, the APIs have been converted to asynchronous with the use of `async/await`, to enhance responsiveness and scalability.

2.5 ReactJS to Dynamic Frontend Apps

ReactJS is a famous JavaScript library to create user interfaces. It allows creation of dynamic and interactive web apps in terms of component-based architecture.

ReactJS has several benefits, such as reusable components, Virtual DOM rendering that is efficient, Easy state management, Easy interconnection with APIs. This internship made use of ReactJS to develop the front-end interface whereby the information in backend APIs was also fetched asynchronously and displayed dynamically.

2.6 PDF to HTML Conversion to process Data

The PDF documents are also commonly used in industries as reporting. But they cannot be directly applied in structured data extraction. It is possible to convert PDF files to HTML format which is more easily parsed and tabular data extracted.

The extraction of data based on HTML is enhanced:

- Data accessibility
- Connections with web applications.
- Automation of report processing.

This was applied within the internship to work on manufacturing reports effectively.

2.7 Speech Recognition Systems (ASR)

Automatic Speech Recognition (ASR) systems convert spoken language into text. Modern ASR systems use deep learning models to achieve high accuracy across multiple languages. In this project, ASR technology was used to recognize speech input in multiple Indian languages, including Telugu, Kannada, Hindi, and Tamil. These systems enable real-time transcription of speech into text, forming the first stage of the multilingual translation pipeline.

2.8 Multilingual Machine Translation Systems

Machine translation systems convert text from one language to another using advanced computational models. Recent developments in transformer-based architectures have significantly improved translation accuracy, especially for low-resource languages. Multilingual translation models enable conversion of text from multiple languages into a target language, such as English. In this project, multilingual translation techniques were used to convert recognized speech text into English, enabling cross-language communication.

Summary

This chapter discussed the key technologies and concepts used during the internship, including full-stack development, REST APIs, asynchronous processing, ReactJS frontend development, PDF-to-HTML conversion, speech recognition, and multilingual machine translation.

These technologies collectively enabled the development of a complete web-based reporting system and a real-time multilingual speech translation application.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter describes the step-by-step methodology followed during the internship at Rail Wheel Factory (RWF). The work was carried out in multiple phases, starting from basic database development, progressing to full-stack application development, and finally extending to an AI-based mini-project. Each phase contributed to building a comprehensive understanding of real-world software development practices.

3.2 Phase 1 – Student Database Training Project

The first step was dedicated to the knowledge of the basics of database design and background development. Activities performed:

- created relational database tables in SQL Server.
- Added and controlled data records.
- Used SQL functions such as: Joins, Filtering, Aggregations
- Developed stored procedures to help in managing data effectively.
- Created REST APIs based on FastAPI with asynchronous support(async/await).
- APIs tested with Postman.
- Implemented frontend using ReactJS (JavaScript and TypeScript)
- Performed CRUD operations: GET, POST, PUT, DELETE

This stage assisted in establishing a good base in full-stack development.

3.3 Phase 2 -RWF Report API Development.

During this stage, emphasis was made on real world industry data. Activities performed

- Breakdown of company site structure and report folders analysed.
- Created APIs of various report files.
- Linked APIs with SQL Server database.
- Installed logic of data retrieval of manufacturing reports. Ensured that there were correct API responses in the form of JSON.

This stage brought about real-world issues of dealing with actual industrial data.

3.4 Phase 3- API Asynchronous Implementation.

Firstly, the APIs were deployed with the help of synchronous processing. But the APIs were transformed into asynchronous APIs so as to enhance performance and scalability. Activities performed:

- APIs refactored with async/await.
- Introduced non-blocking request service.
- Better large data response time.
- Enhanced backend performance

This stage enhanced the efficiency of the system, and allowed managing multiple requests at the same time.

3.5 Phase 4- PDF to HTML Conversion and Data Processing.

The industrial reports were in PDF format thus had to be preprocessed to integrate into the web. Activities performed:

- Downloaded and analyzed PDF reports files.
- Created PDF documents that were transformed into HTML.
- Extracted structured information in HTML.
- APIs extracted data processed with Flask.
- Improved processed data into the system.

The phase allowed the generation of reports dynamically and a web-based display.

3.6 Phase 5 – Frontend Integration

ReactJS was used to create the frontend which can show the data dynamically. Activities performed:

- Created React components that were reusable.
- backend API integration with frontend.
- Made API calls using used asynchronous fetch (async/await).
- Managed application state
- Furnished real time data on UI.
- Error conditions and loaded hand. This step completed the entire stack integration.

3.7 Phase 6 - Remaining Modules completed.

During the last step of the internship activity:

- Finished left over report files.
- Finalized API integrations
- Assured frontend interoperability of each module.
- Carried out testing and debugging.

This guaranteed the functionality of the system.

3.8 Phase 7 – Mini Project: Multilingual Speech-to-Text

Translation System

In addition to the main internship work, a mini-project was developed to explore AI-based applications.

System Workflow:

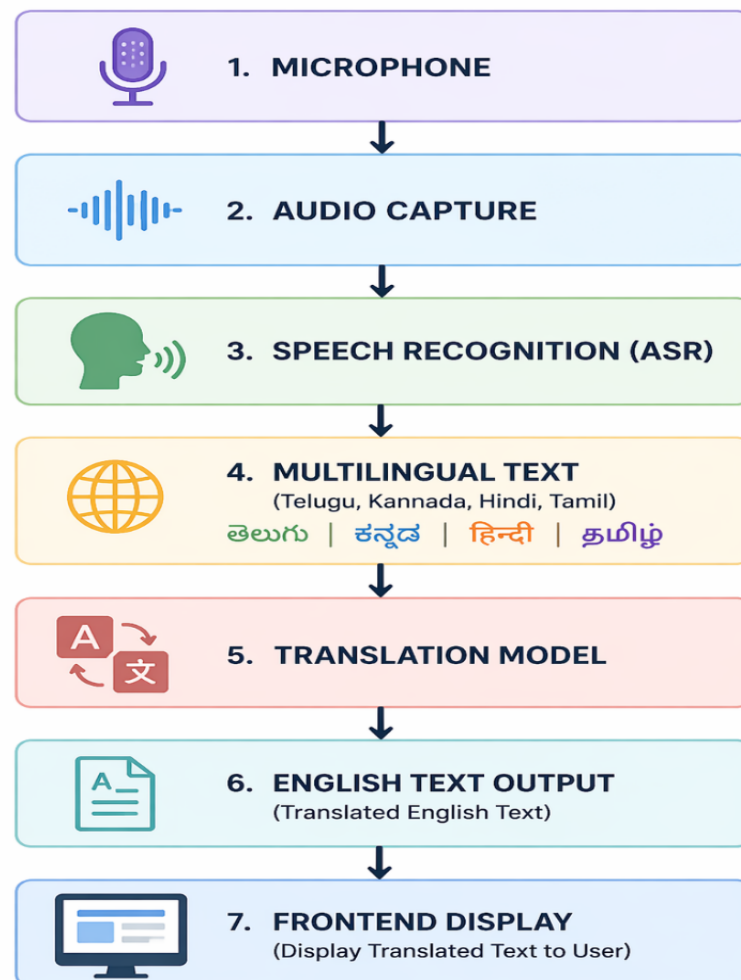


Figure 3.1: Speech-to-Text and Translation Processing Pipeline

Activities Performed:

- Captured live audio input using a microphone
- Processed audio data using appropriate libraries
- Performed multilingual speech recognition (Telugu, Kannada, Hindi, Tamil)
- Translated recognized text into English using multilingual models
- Displayed translated output on the frontend
- Implemented continuous real-time processing

This phase demonstrated the integration of speech recognition and machine translation in a real-time application.

Summary

The methodology followed during the internship covered the complete software development lifecycle, including database design, backend API development, asynchronous processing, data extraction from reports, frontend integration, and AI-based multilingual translation.

This structured approach enabled the successful development of a full-stack reporting system along with an intelligent speech translation application.

Chapter 4

PROJECT MANAGEMENT

4.1 Internship Timeline

The internship work was done in several stages throughout the semester and each stage had certain tasks and goals to be accomplished. The following is a timeline on the course of work throughout the internship.

Table 4.1 Internship timeline

Phase	Phase time/period Work Done
Phase 1	Week 1-2 Week 3 Student database project, SQL operations, CRUD API
Phase 2	Week 3-4 Development of APIs in RWF report files.
Phase 3	Week 5-6 Conversion of APIs synchronous to asynchronous.
Phase 4	Week 7-8 to PDF to HTML conversion and data extraction
Phase 5	Week 9-10 React frontend development and implementation
Phase 6	Week 11-14 Completion of remaining modules, testing, and development of multilingual speech translation mini-project

This incremental process allowed a consistent growth and understanding throughout the process.

4.2 Risk Analysis

There were several operational risks and technical risks which were identified during the internship. These risks were addressed with help of appropriate strategies. Risks and Mitigation identified: Risk Description Mitigation Strategy. API Failure Backend API cannot respond to incorrect queries or fails because of server problems Implemented error handling

and testing with Postman. Data Inconsistency Wrong or redundant information in database Used validation and orderly queries. Performance Problems Slower APIs due to synchronousness Turned previously asynchronous APIs into synchronous. PDF Parsing Processes Could not get structured data out of PDF easily Converted to HTML to easier parse. Frontend Integration Issues Data not rendering properly Debugging and proper state management in React System Crashes Runtime error errors Implemented exception handling and testing. This analysis was effective in raising the problem areas in advance and a smooth development of the system.

4.3 Internship Budget

This internship project was mainly software based and needed not much financial input. Budget Details: Resource Description Cost Hardware Personal laptop to develop on Existing. Software Python, Flask, ReactJS, SQL Server Free/ Open-source. VS Code Postman, GitHub Free. Internet API access and research needed Minimal.

Summary: The project was economically friendly, most of the tools were open-source, no extra (infrastructure) expense was involved.

Summary

Project management approach made sure that the work done in the internship was done in a systematic and effective nature. This schedule was very useful in monitoring the process, risk analysis guaranteed stability, and the low budget projected cost effectiveness.

Chapter 5

ANALYSIS AND DESIGN

5.1 System Requirements

The system requires both functional and non-functional components to ensure proper operation.

Functional Requirements

- The user should be able to view manufacturing reports
- The system should retrieve data through backend APIs
- APIs should fetch data from SQL Server
- The system should process PDF reports into structured format
- The frontend should dynamically display data
- The system should support real-time multilingual speech-to-text translation

Non-Functional Requirements

- High performance using asynchronous APIs
- Scalability for future expansion
- Reliable and consistent data processing
- User-friendly interface
- Efficient handling of multiple requests

5.2 Block Diagram

The system would have a stack of three layers:

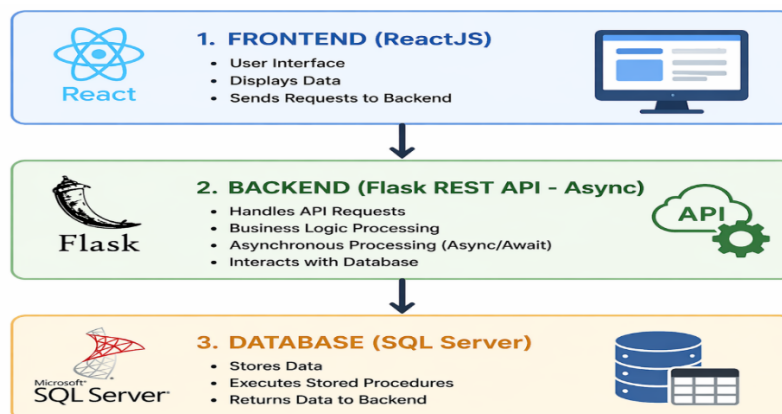


Figure 5.1: Architecture of React–Flask REST API with SQL Server Backend

This modular design guarantees the separation of concerns and maintainability.

5.3 System Flow Chart

There are the following flow system process reports:

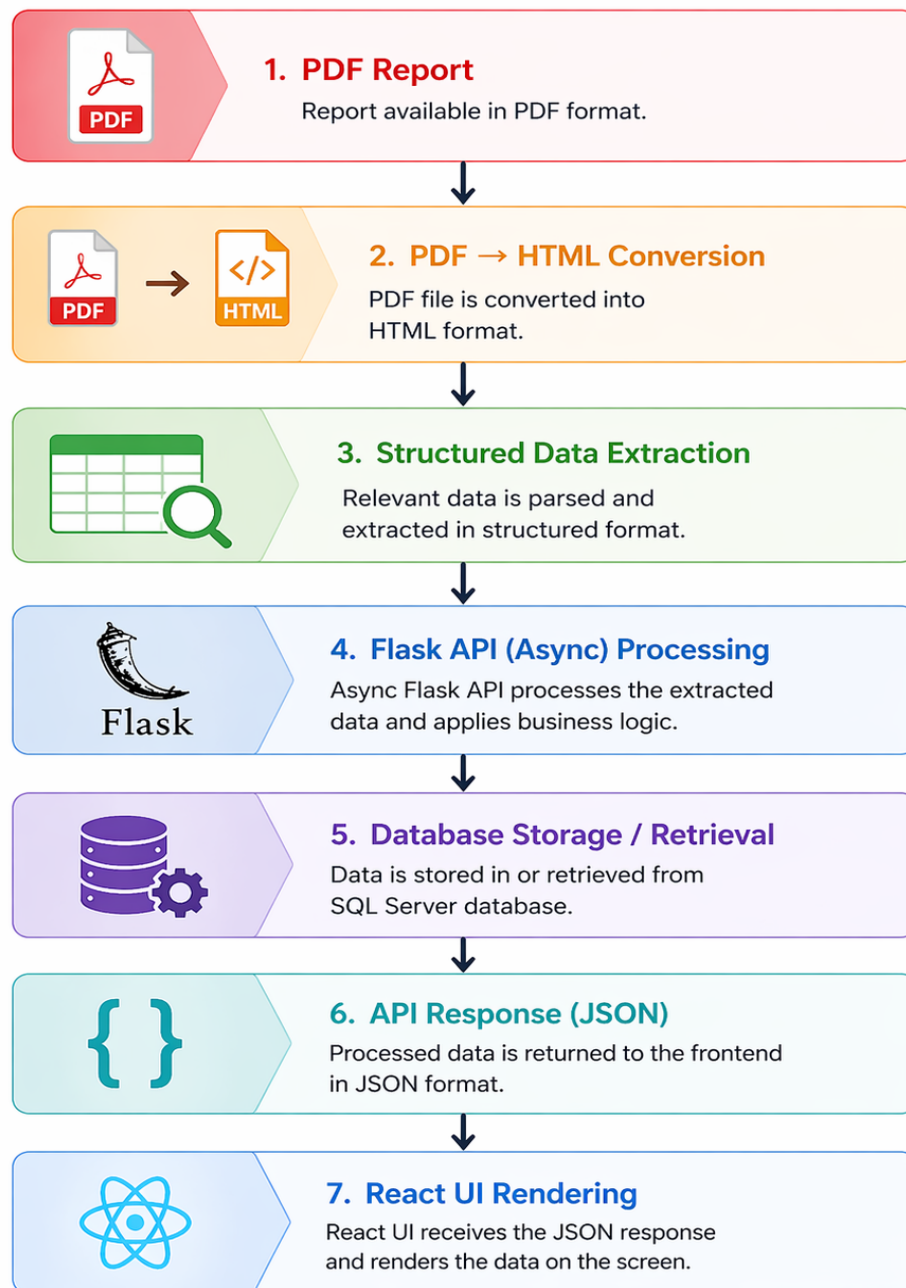


Figure 5.2: End-to-End Data Processing Pipeline (PDF to React UI)

This flow facilitates the dynamic generation of reports out of the PDFs which have been made up of.

Mini Project Flow:

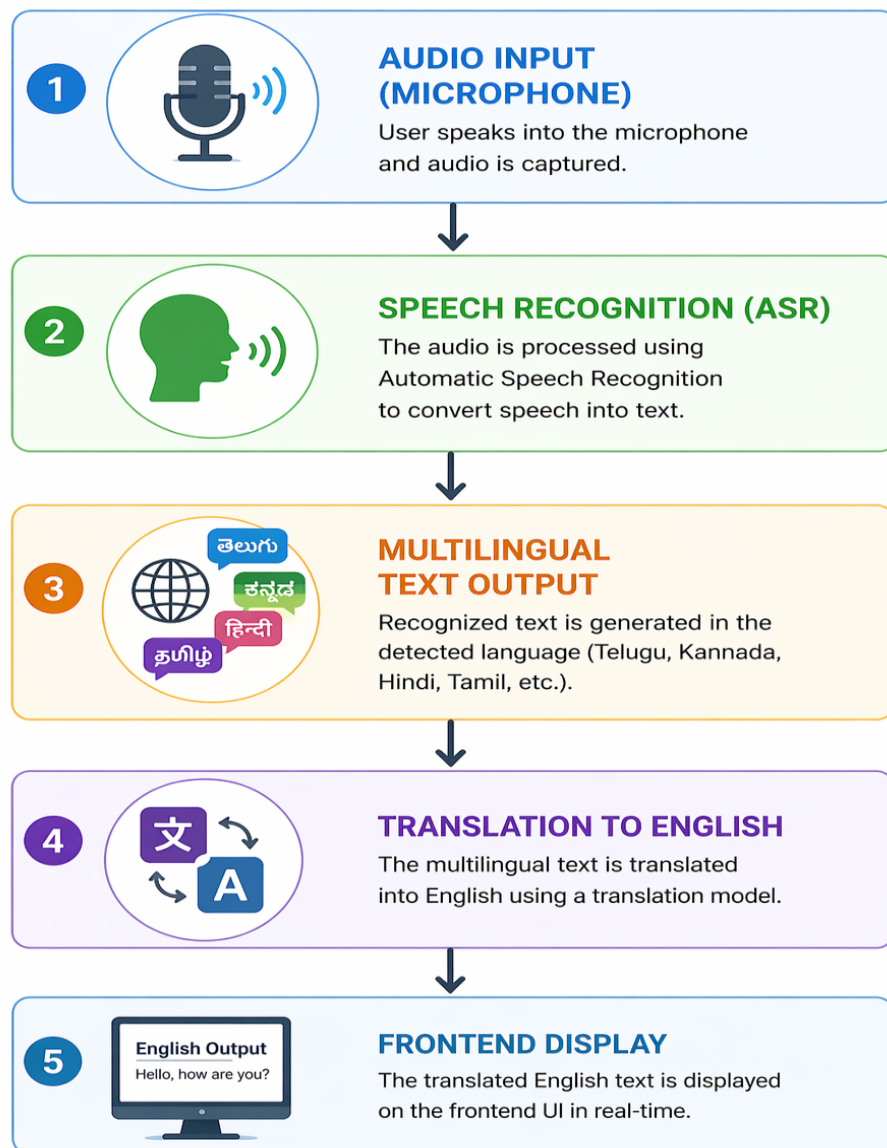


Figure 5.3: Workflow of Real-Time Speech-to-Text and Translation System

5.4 Database Design

To store structured data the database was designed on SQL Server.

Key Features:

- Relational tables Use of primary and foreign keys.
- Stored procedures to have an efficient data retrieval.
- Streamlined queries that are more efficient.

The database maintains information about the report data, system records and information that has been processed.

5.5 API Architecture Design

The REST API (FastAPI) was used to create the backend.

Key Design Features:

- RESTful API endpoints
- HTTP methods (GET, POST, PUT, DELETE) are used.
- JSON-based communication
- Checking and correcting errors.
- Asynchronous implementation with async/await.

This architecture provides effective communications between frontend and backend.

5.6 Frontend Design

The reactjs was used to build a frontend.

Features:

- Component-based architecture
- Dynamic data rendering

Asynchronous API calls from fetch.

- React hook management of states.
- Responsive UI design the frontend corresponds with APIs and shows real-time information.

5.7 System Integration

The system comprises several integrations: Database layer (SQL server) Back-end layer (Fast APIs), Frontend layer (ReactJS) Components are all connected so that they offer end to end functionality, AI components (ASR + Translation models).

Summary

This chapter presented the complete system design, including architecture, workflow, database structure, API design, frontend implementation, and multilingual speech translation pipeline. The modular design ensures scalability, efficiency, and ease of maintenance.

Chapter 6

SOFTWARE TOOLS

6.1 Software Development Tools

Development, testing, and deployment were done by various software tools and technologies. Backend Development - Python as the main program language, Fast API Framework application to develop REST APIs, request-response communication is dealt.

Frontend Development - ReactJS used to create dynamic user interfaces, JavaScript fratricious to logics on the frontend, TypeScript type-safe frontend development is used, Database SQL Server Used in storing structured information and handling, Relational database operations with support.

API Testing - Postman used to test API endpoints validated request-response behavior Development Environment, Visual Studio Code Developed code editor, GitHub Used for version control and code management.

AI and Mini Project Tools

sounddevice Library: Used for capturing real-time audio input from the microphone

ElevenLabs Scribe v2: Used for multilingual speech recognition (Telugu, Kannada, Hindi, Tamil)

Meta NLLB-200 Model: Used for multilingual translation (speech text → English)

Hugging Face Transformers: Used for loading and managing translation models

6.2 Testing Environment

The system has been tried on a local development environment.

Testing Methods - Postman API testing, Frontend testing using browser (Chrome/Edge), Console log and error handling debugging, Frontend/backend integration testing, Live audio test a speech translation pipeline using live audio as input.

6.3 Mini Project Testing (Multilingual Speech Translation)

The mini-project was tested using real-time audio input across multiple languages.

Testing Approach:

1. Captured live speech input using a microphone

2. Tested speech recognition for:
 - Telugu
 - Kannada
 - Hindi
 - Tamil
3. Verified accuracy of recognized text
4. Evaluated translation output in English
5. Displayed results on frontend in real time

This ensured that the system could handle multilingual speech input and provide accurate translation output.

Summary

This chapter described the software tools and technologies used during the internship. The use of modern development frameworks such as FastAPI and ReactJS enabled efficient full-stack development, while the integration of AI tools supported the implementation of a real-time multilingual speech translation system.

Chapter 7

EVALUATION AND RESULTS

7.1 Test Points

The system was tested at various levels to check the accuracy, performance and stability. Key Test Points:

- Connection to database and execution of queries.
- API response correctness
- Asynchronous API performance
- Accuracy of PDF to HTML conversion.
- Report interpretation of data.
- Frontend data rendering
- Connection between front and back.
- Speech recognition accuracy
- Translation output correctness

7.2 Test Plan

Each functionality of the system was tested using a structured method to confirm it. Testing Strategy:

- Unit Testing: Single components (APIs and database queries), were tested independently.
- Integration Testing: Authenticated communication among frontend, backend and database.
- System Testing: Validated the entire process taking input up to output.
- Real-time Testing: Cross validated speech translation system with real-time microphone speech input.

7.3 Test Results

All modules underwent testing and were successful. API Testing REST APIs: all APIs responded with the appropriate amount of correct JSON. CRUD operations: No issues. Asynchronous APIs enhanced the response time. There were numerous requests being served by APIs.

Frontend Testing: Information was brought in and properly presented in React UI,

asynchronous fetch calls did not have page reload, loading and error states handled properly, UI presented organized data of reports in a lucid manner. PDF Processing Results Conversion to HTML of PDF files was successful, organized data mining obtained, correct data processing and displaying of report data.

7.4 Insights

The implementation and testing provided the insights as follows: Asynchronized APIs helped a lot in improving the performance of the system.

- Conversion of PDFs to HTML made processing of data easy.
- ReactJS was used to deliver a dynamic user interface which was responsive.
- End-to-end integration was provided to make sure that the information flows seamlessly.
- Multilingual speech recognition and translation demonstrated practical real-time AI application

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1 Social Aspects

The system developed also gives positive contributions to the industrial and society needs through increasing efficiency and lessening the amount of manual work. Minimizes human effort to generate reports. Improves accuracy of data handling. Facilitates quicker choices in the industrial processes. Increases the availability of information with web-based systems.

8.2 Legal Aspects

The system adheres to the general software and data usage. The data processed is associated with in-organizational use. None of the sensitive information access without permission. APIs are made to be used in a controlled manner.

8.3 Ethical Aspects

The ethics were taken care of throughout development. Ensured correct representation of data, did not or did not misuse user or organization information, was transparent on system functionality.

8.4 Sustainability Aspects

The system promotes sustainable practices. Digitizes paper-based reporting. Minimizes resource consumption. Promotes effective data management. Promotes scalable solutions in the long-term.

8.5 Safety Aspects

Software usage is safe in the system.

- Effective management of errors to prevent system crash.
- Handling of data is secure.
- Controlled API responses

Summary

In this chapter the authors pointed out the social effect, legality, ethical considerations, sustainability, and safety concerns of the developed system. The system facilitates responsible, efficient, and secure industrial operations.

Chapter 9

CONCLUSION

The trainee experience at Rail Wheel Factory (RWF) was a good practical work in full-stack development and in practice of the real-life industry. This task covered the design and development of a web-based reporting system based on SQL server and Flask REST API and ReactJS.

The internship started with basic training via a project with student database, then development of APIs with real industrial data. This was further accomplished by making synchronous APIs asynchronous, which increased performance and scalability of the system. The processing of PDF reports into HTML format and extraction of structured information to be displayed in a dynamic web were effective.

In addition to the core internship work, a mini-project on real-time multilingual speech-to-text translation was developed. The system supports multiple Indian languages, including Telugu, Kannada, Hindi, and Tamil. It captures live audio input, performs speech recognition, and translates the recognized text into English using advanced AI models. The translated output is displayed on the frontend in real time, demonstrating an end-to-end multilingual communication pipeline.

Overall, the internship significantly enhanced technical skills in backend development, frontend integration, asynchronous processing, and AI-based systems. It also provided valuable exposure to industrial workflows and real-world problem-solving. The experience gained during this internship forms a strong foundation for future professional growth in software development and data-driven applications.

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APPENDIX – A

API IMPLEMENTATION (PSEUDOCODE)

Define API endpoint for report retrieval

Receive input parameters (e.g., date)

Call database stored procedure

Fetch data from SQL Server

Process and format data as JSON

Return response to frontend

APPENDIX – B

FRONTEND API CALL

Send request to backend API

Receive JSON response

Store data in state

Render data dynamically in UI

Handle loading and error states

APPENDIX – C

MULTILINGUAL SPEECH TRANSLATION WORKFLOW

Capture audio input from microphone
Convert audio to required format
Perform speech recognition (multi-language)
Generate text output
Translate text to English
Display result on frontend

APPENDIX – D

Figure D.1 - Output

